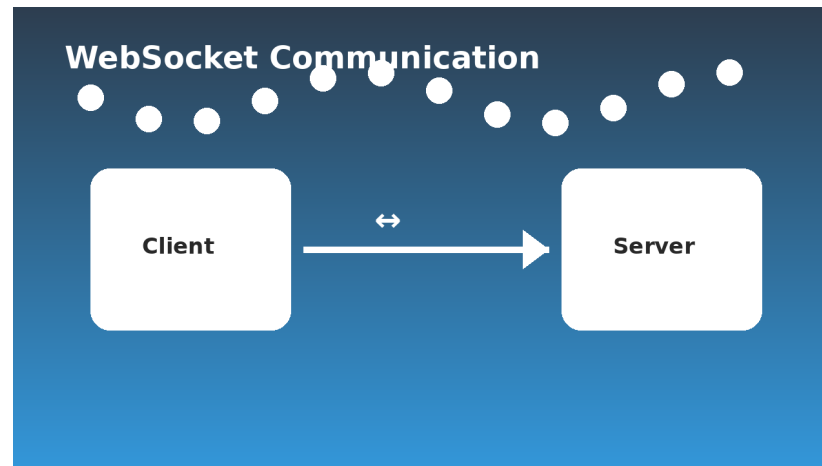


WebSockets

Prepared By: Dr. D. P. Singh

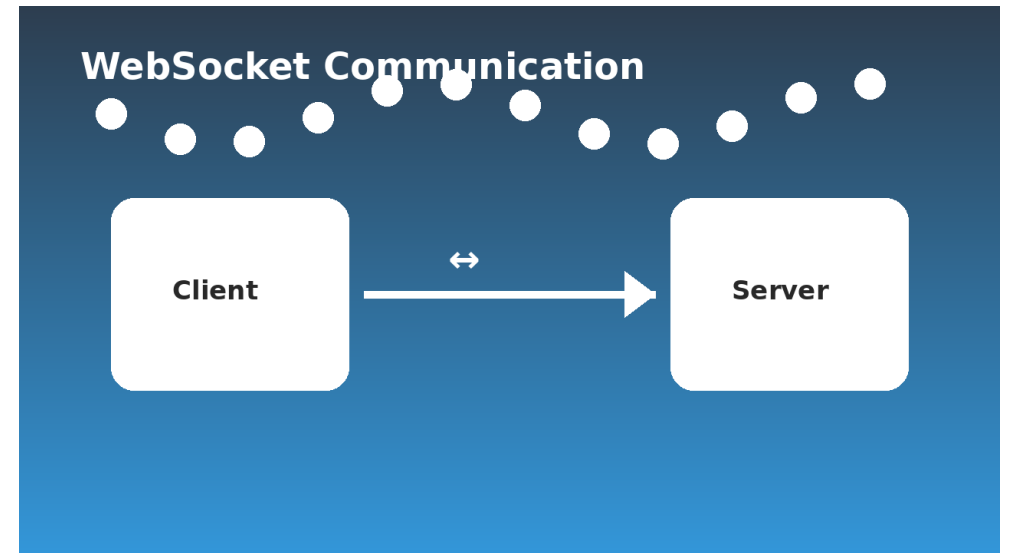
WebSockets Overview

- WebSockets provide **full-duplex communication** between client and server.
- They maintain a **persistent connection** after the initial handshake.
- Useful for real-time applications like **chat apps, gaming, and live updates**.



How WebSockets Work

- Client **initiates** a WebSocket handshake using **HTTP**.
- Server **upgrades** the connection to **WebSocket protocol**.
- After upgrade, data flows continuously in **both directions**.



Features of WebSockets

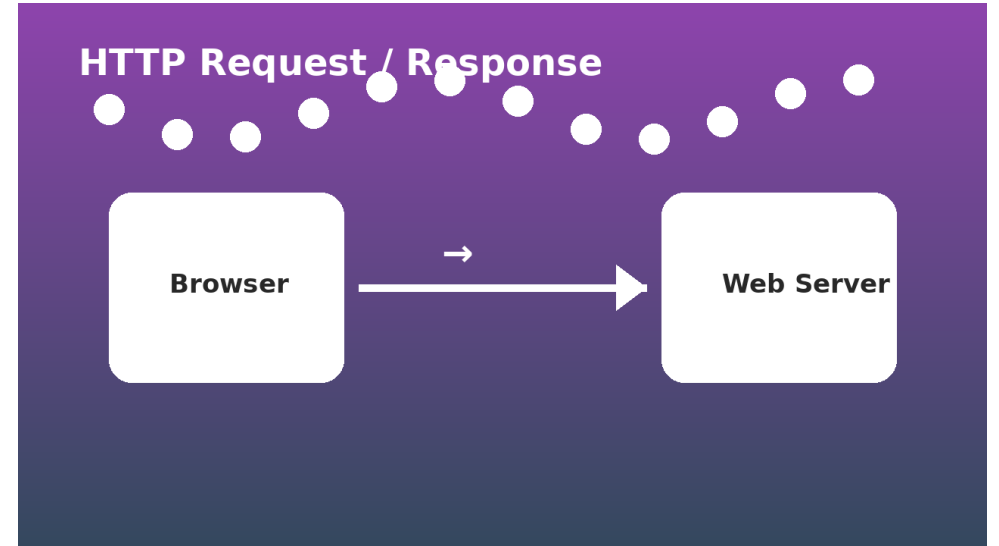
- Low latency communication
- Persistent connection
- Efficient for frequent data exchange
- Supports real-time updates

Applications of WebSockets

- Live chat systems
- Stock market updates
- Online multiplayer games
- Collaborative editing tools

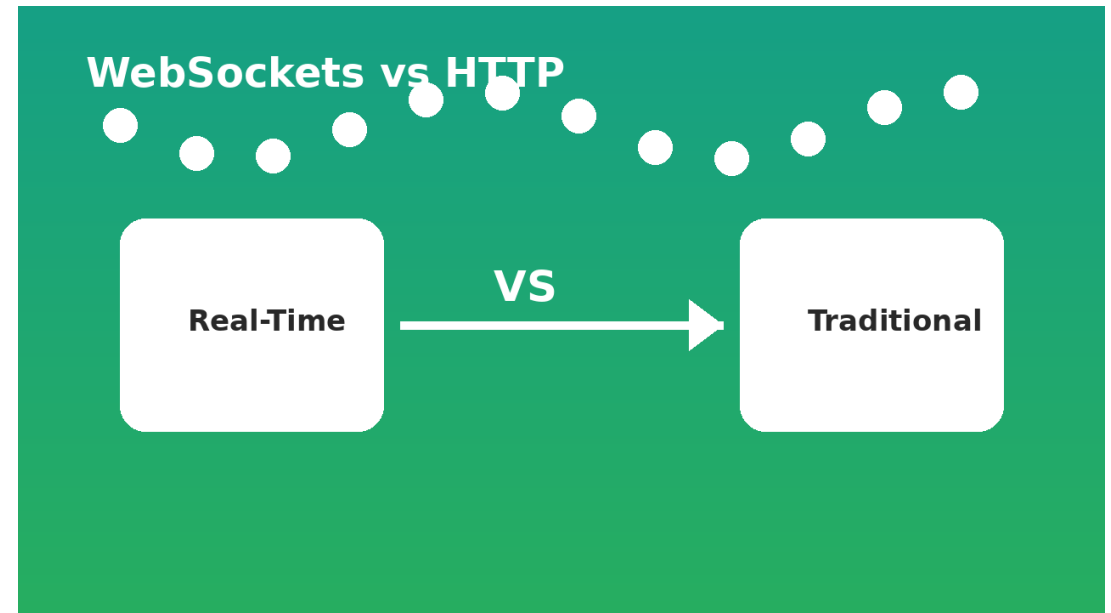
HTTP Overview

- HTTP is a **request-response protocol**.
- Client sends a request, server sends a response.
- Connection is usually **closed after response**.



WebSockets vs HTTP

- WebSockets are full-duplex; HTTP is half-duplex.
- WebSockets maintain persistent connections; **HTTP is stateless**.
- WebSockets are better for real-time apps; HTTP is ideal for standard web content.



Comparison Table

- Connection: Persistent (WebSockets) vs Temporary (HTTP)
- Communication: Bidirectional vs Request-Response
- Latency: Low vs Higher
- Use Cases: Real-time apps vs Traditional websites

- Thank You